



Neural Networks & Deep Learning

1.Introduction

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Vatican Secret Archives

A 1,000-Year-Old Mystery

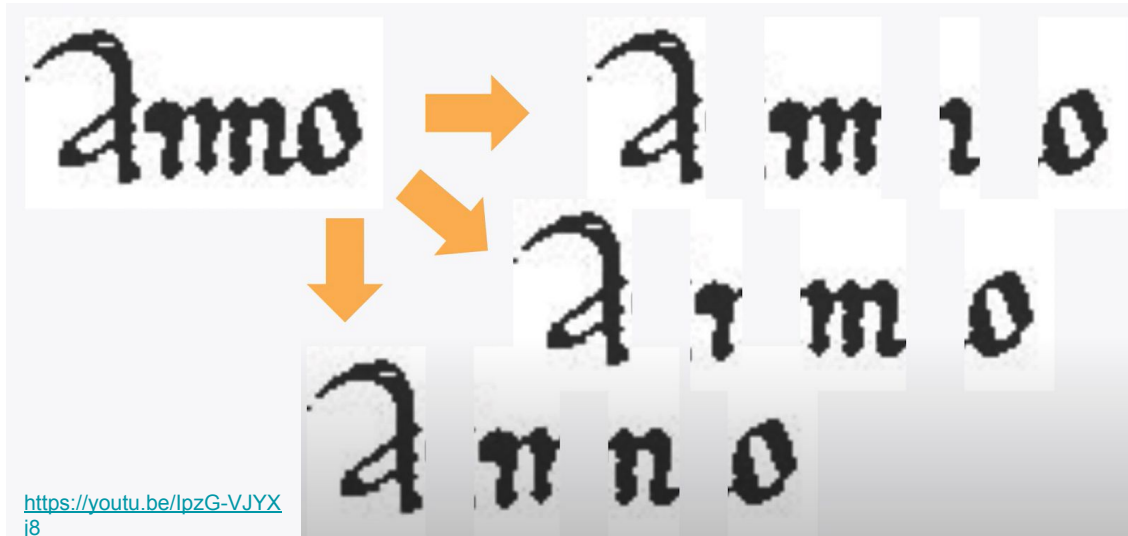
- The Vatican Apostolic Archives: 85 kilometers of shelving containing millions of medieval manuscripts.
- The Problem: Written in "Mercantesca" (a medieval script) that is nearly impossible for standard computers to read.
- The Challenge: Even experts take years to transcribe a single volume. We need a way to automate this at scale.



Vatican Secret Archives

Why Traditional OCR Failed?

- Traditional OCR: Tries to recognize characters by their rigid shapes.
- The Complexity: In medieval Latin, characters are "connected" (ligatures) and handwriting varies wildly between scribes.



Vatican Secret Archives

The Deep Learning Solution (In Codice Ratio)

- Crowdsourcing: Developed an app to identify individual letters.
- Synthetic Data Generation. Researchers took these individual characters to create millions of synthetic sentences.
- These synthetic sentences (where the computer already "knows" the answer) were used to train a Deep Neural Network.
- The Result: By practicing on "fake" digital sentences made of real character images, the AI learned to read the actual, ancient manuscripts.

*“In the not-so-distant future,
anyone with an idea and
basic coding skills will be
able to build smart
applications that learn from
data.”*

François Chollet
Creator of Keras library



DEEP LEARNING with Python

THIRD EDITION

François Chollet
Matthew Watson

 MANNING

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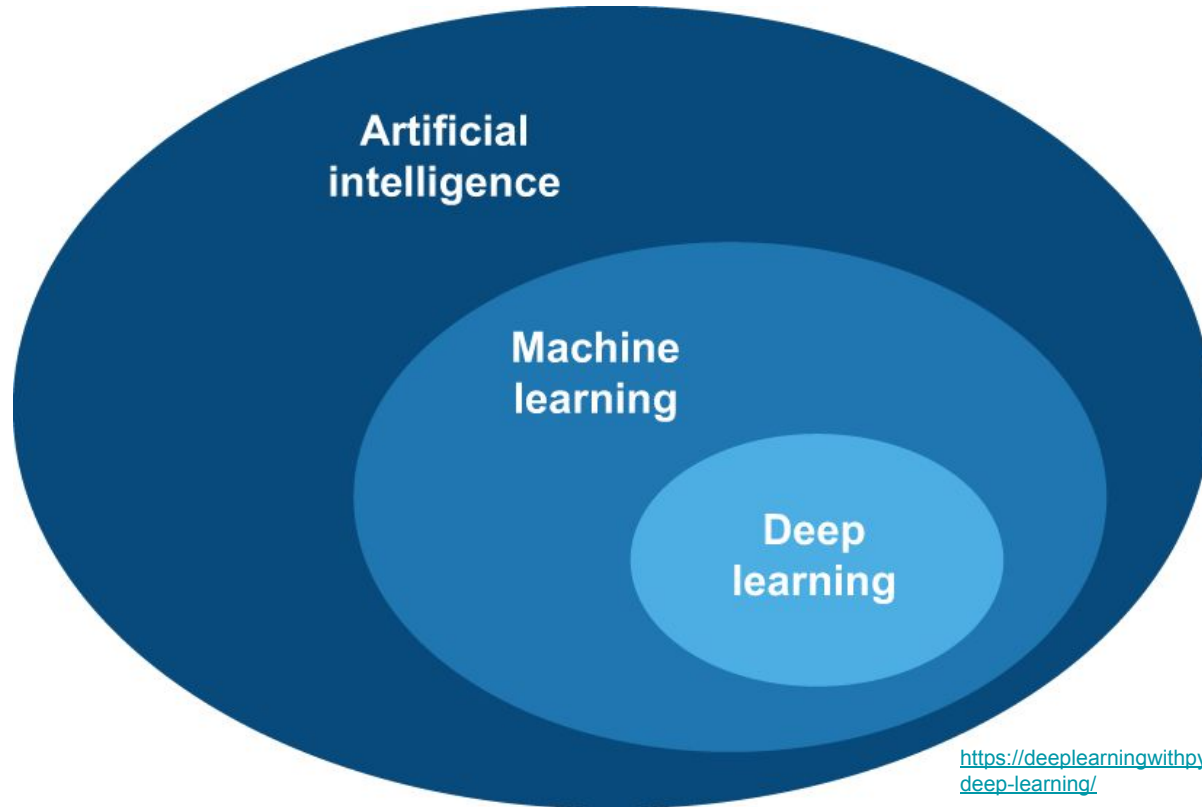


Perfect for anyone interested in learning by doing from one of the industry greats.

— Anthony Goldbloom, Founder of Kaggle



AI vs. Machine Learning vs. Deep Learning



Artificial Intelligence

What is Artificial Intelligence?

- **Core idea:** Any technique that enables computers to mimic human behaviour
- A Broad Field: It encompasses Machine Learning and Deep Learning.
- Beyond Learning: It also includes many approaches that do not involve any learning.
 - e.g., Early chess programs used hardcoded rules and did not qualify as machine learning.

Artificial Intelligence

Symbolic AI

- The main approach in Symbolic AI is to use logic-based programming, where rules and axioms are used to make inferences and deductions.
- Programmers manually "craft" a large set of explicit logical rules to solve problems.
 - E.g., *IF patient has a fever AND patient has a cough AND patient has difficulty breathing THEN patient may have pneumonia.*

Artificial Intelligence

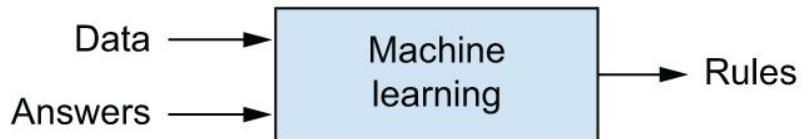
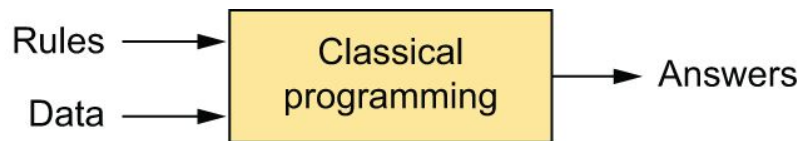
The Success and Failure of Symbolic AI

- Symbolic AI is very effective for well-defined logical problems (e.g., playing chess)
- However, it is nearly impossible to create explicit rules for "fuzzy" problems.
- Fuzzy Problems
 - Image classification
 - Speech recognition
 - Natural Language Translation
- Because rules couldn't be written for these complex tasks, Machine Learning arose to take its place.

Machine Learning

A New Paradigm for Computing

- Classical Programming: A human programmer writes explicit Rules (a program) to turn Data into Answers.
- Machine Learning: The process is turned around—the machine looks at Data and Answers to figure out what the Rules should be.
- **The Shift:** We move from "explicitly programming" a system to "training" it.



Machine Learning

How Machine Learning "Learns"

- **Core idea:** Ability to learn without explicitly being programmed
- **Training:** The system is presented with many examples relevant to a specific task.
- **Statistical Structure:** The algorithm searches for patterns or "statistical structure" within these examples.
- **Automation:** These discovered patterns eventually allow the system to create the rules needed to automate the task.

Machine Learning is a hands-on discipline driven by practical findings and advances in hardware and software.

Machine Learning

The Three Necessities for Machine Learning

- **Input Data Points:** The raw information (e.g., sound files for speech recognition or pictures for image tagging).
- **Examples of Expected Output:** The "labels" or targets we want the model to predict (e.g., human-generated transcripts or tags like "dog" and "cat").
- **A Measurement of Performance:** A way to calculate the "distance" between the machine's current guess and the true answer. The measurement is used as a feedback signal to adjust the way the algorithm works. This adjustment step is what we call learning.

Machine Learning

Supervised Learning

Data: (x, y)
 x is data, y is label

Goal: Learn function to map

$$x \rightarrow y$$

Example:



This thing is a bicycle

Unsupervised Learning

Data: x
 x is data, no labels

Goal: Learn underlying structure

Example:

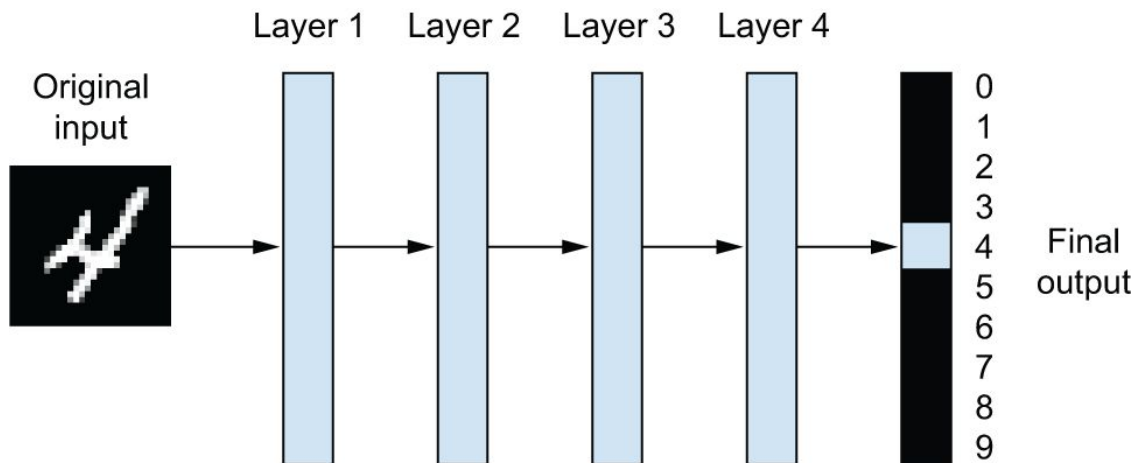


This thing is like the other thing

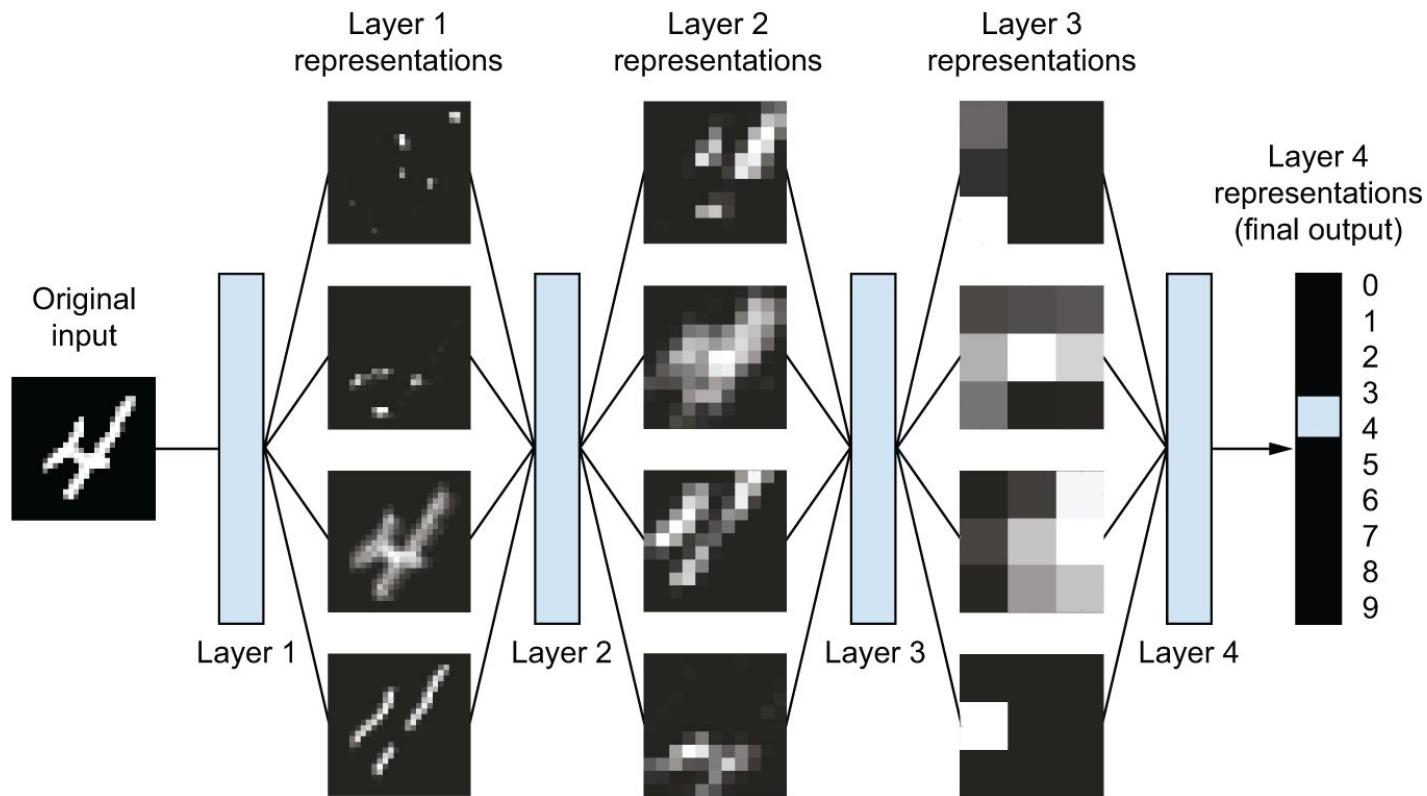
Deep Learning

Defining "Deep" Learning

- The "deep" in deep learning does not refer to a "deep understanding" of the data.
- It refers to the use of successive layers of representations.
- The number of layers contributing to a model is known as its depth.
- It can be thought of as "layered representations learning" or "hierarchical representations learning"



Deep Learning



Deep Learning

Deep Learning vs. Machine Learning

Feature	Traditional Machine Learning	Deep Learning
Layers	Focuses on only one or two layers of representations	Often involves tens or hundreds of successive layers.
Automation	Often requires manual feature engineering.	Layers are learned automatically from exposure to training data.
Terminology	Sometimes referred to as shallow learning	A multistage mathematical framework for learning representations

Deep Learning

Misconception: The Brain Metaphor

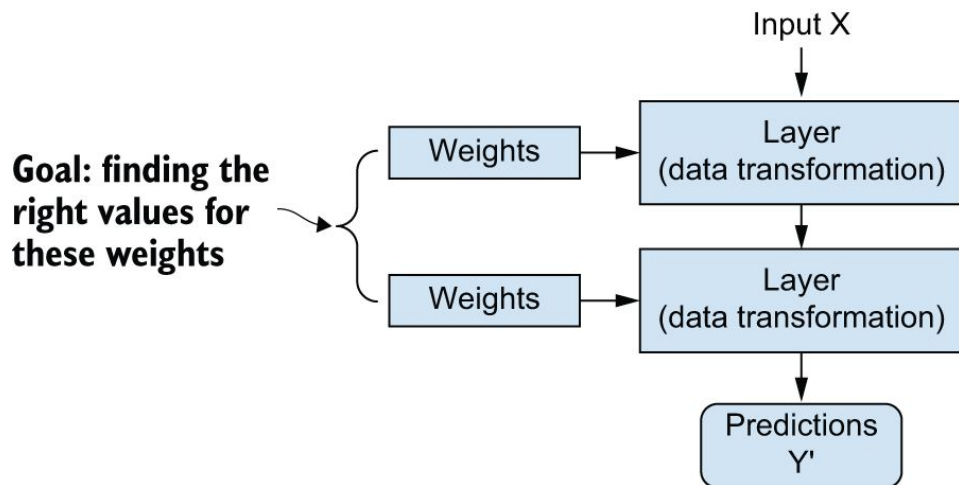
- While inspired by neurobiology (specifically the visual cortex), deep learning models are not models of the brain.
- There is no evidence that the brain uses learning mechanisms similar to those in modern deep learning.
- It is counterproductive to think of deep learning as "just like our minds"
- **The Bottom Line:** Deep learning is a mathematical framework, not a biological one.

Example analogy: A Boeing 747 was inspired by birds (wings, aerodynamics), but a 747 does not flap its wings, and it doesn't have feathers

Deep Learning

Understanding how deep learning works

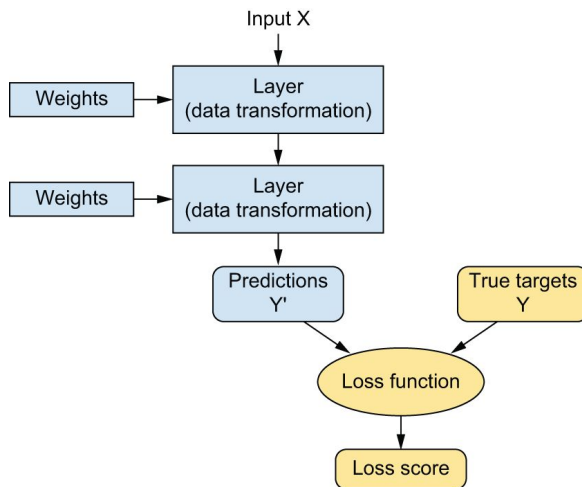
- The specification of what a layer does to its input data is stored in the layer's weights, which in essence are a bunch of numbers.
- In technical terms, we'd say that the transformation implemented by a layer is parameterized by its weights



Deep Learning

Understanding how deep learning works

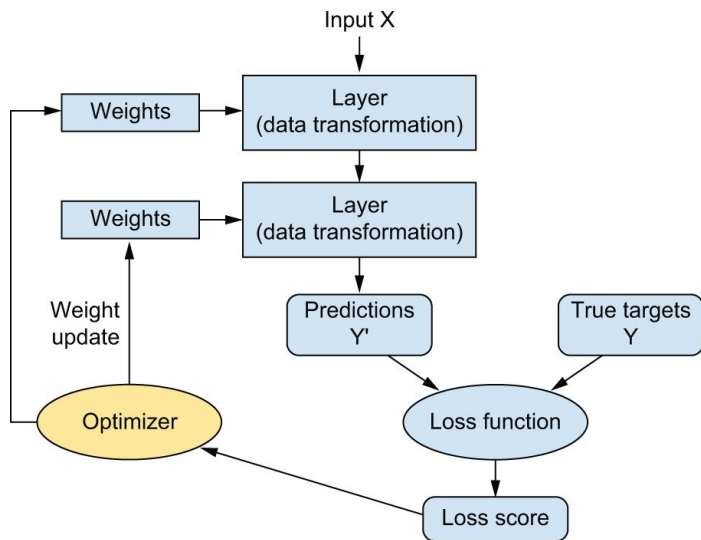
- To improve a network, we must first measure how far its output is from the expected result.
- This is the job of the loss function of the network, also sometimes called the objective function or cost function.



Deep Learning

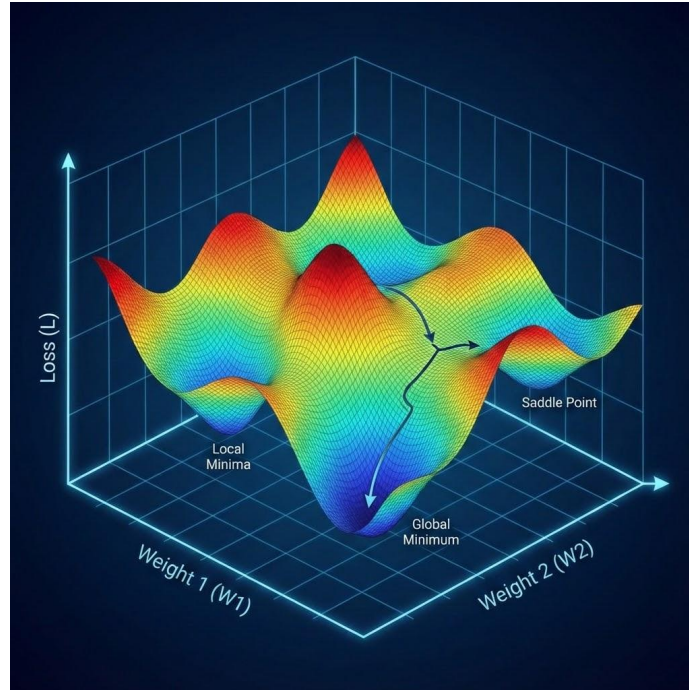
Understanding how deep learning works

- Deep learning uses the Loss Score as feedback to adjust the weights slightly
- This component implements the Backpropagation algorithm, which is the "brain" of the learning process.



Deep Learning

Understanding how deep learning works



What makes deep learning different

- **Simplicity**

- Automate feature engineering
- This replaces complex, multi-step systems with one simple "end-to-end" model.

- **Scalability**

- Deep learning is designed to run on GPUs and specialized chips. It gets faster as computer hardware improves
- These models learn by looking at data in small batches. This means they can be trained on datasets of any size, no matter how huge.

- **Versatility and reusability**

- Deep learning models can be trained on additional data without restarting from scratch. This making them viable for continuous online learning - an important property for very large production models.
- You can take a model that already "knows" the world and apply it to a new job with almost no extra training (e.g., Once a model is trained to "understand" human language (like a Foundation Model), you can use it to write poems, summarize legal papers)

What deep learning has achieved so far

- Fluent and highly versatile chatbots such as ChatGPT and Gemini
- Programming assistants like GitHub Copilot
- Photorealistic image generation
- Human-level image classification
- Human-level speech transcription
- Human-level handwriting transcription and printed text transcription
- Dramatically improved machine translation
- Dramatically improved text-to-speech conversion
- Human-level autonomous driving, already deployed to the public in Phoenix, San Francisco, Los Angeles, and Austin as of 2025
- Improved recommender systems, as used by YouTube, Netflix, or Spotify
- Superhuman Go, Chess, and Poker playing

References

- Main text + images used for the lecture is based on :
<https://deeplearningwithpython.io/>
- <https://developer.nvidia.com/blog/ai-helps-researchers-unlock-mysteries-of-vatican-archives/>
- <https://picampus-school.com/in-codice-ratio/>
- Deep learning playlist by MIT :
https://www.youtube.com/watch?v=alfdI7S6wCY&list=PLtBw6njQRU-rwp5_7C0oIVt26ZgjG9NI

Questions & Answers



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